

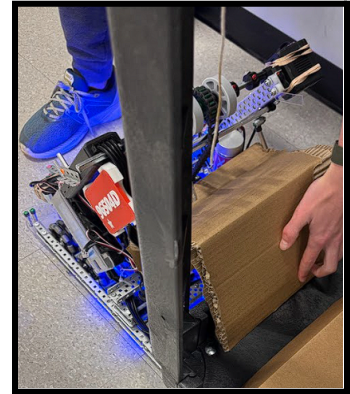
Element G: Construction of a Testable Prototype

The four-wheeled, motor-powered self-driving or remotely controlled vehicle (we have not decided yet) colloquially known as the “boombot” addresses Mr. Kiser’s difficulty pushing the cart due to his back pain and the low handle position requiring him to bend over. The design minimizes the force he needs to apply by providing forward propulsion.

Still, key challenges include ensuring structural strength under significant pressure, achieving sufficient wheel traction for ground stability, and delivering enough motor strength to pull the cart effectively. These issues are addressed through optimized weight distribution, durable materials, and a motor capable of generating the necessary torque.

Initially, we used Cole’s VEX robot to gauge the project's plausibility. After creating a rudimentary attachment system (Fig 1), we saw that 6 (11W 2.1N*m 2.5A) motors were able to easily pull the cart with a 3d printer.

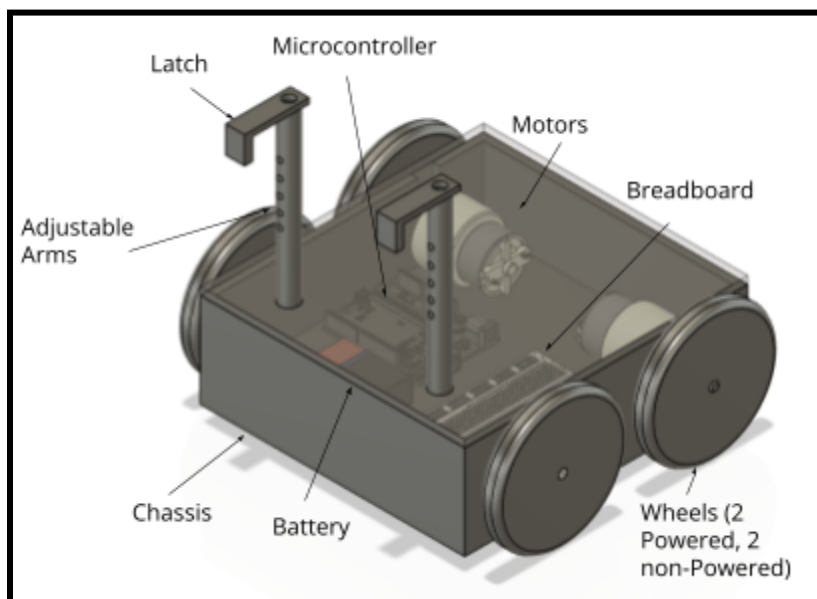
After finding some initial success, we used PVC, wood, and screws to construct a first prototype. We started with the clamp system, designing two PVC hooks that “grab” the bottom ledge of the cart. Then, we connected the hooks to a piece of wood with two caster wheels attached to the bottom.



One drawback of the rudimentary prototype was that it tended to fall forward because the center of mass was behind the wheels, instead of over them. To help fix this, we added more wood to bring the center of mass closer to the front of the prototype, as well as added a third caster wheel for stability.



With our initial prototype completed, we created a CAD in Fusion 360 that was much more accurate to our solution, with a full chassis, microcontroller, motors, battery, and wheels. This allowed us to model some of the more complex aspects of the robot, and create a layout for the parts that we currently don't have the budget to purchase for prototyping.



Between our physical and theoretical models, we should be able to test each of the following subsystems:

Wheels: Four large circular components on the sides, providing movement and traction. Through the CAD, we can test different materials for traction and support via 3D printing.

Motors: Cylindrical objects near the non-clamp side of the robot, responsible for powering the wheels. We could use the CAD and product details of the motors we choose to combine the stall torque, stall current, and wheel size to calculate the pulling power of the pod.

Battery: Red-striped rectangular block in the back section, supplies energy to the motors and microcontroller. We could use the motor product details to calculate the power draw and use the timing data recorded from the testing trip length to make sure the battery capacity is sufficient.

Clamp Arms: Upright structures extending from the top, designed to connect the boombot to the cart. We could test different clamp solutions using the PVC pipes and hooks on the physical solution.

Chassis: The overall body of the robot. The weight of the chassis as a whole appears to be part of the equation to determine how much weight the pod will be able to pull. This is something that we will need an expert opinion on, so we will discuss it with our primary stakeholder and physics teacher, Mr. Kiser.